NARASARAOPETA
ENGINEERING COLLEGE
(AUTONOMOUS)

Department of Artificial Intelligence & Machine Learning

II Year II Semester

FULL STACK DEVELOPMENT-1 LAB

(R23CC22L2)

Roll Number: ________________

Name of the Student: ______________________

Academic Year: 2024-2025

NARASARAOPETA
ENGINEERING COLLEGE
(AUTONOMOUS)

Narasaraopeta Engineering College

Kotappakonda Road, Yellamanda (Post), Narasaraopet – 522601, Guntur District, AP

Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada, Code: 47,

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NARASARAOPETA ENGINEERING COLLEGE
(AUTONOMOUS)

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Kotappakonda Road, Yellamanda (V), Narasaraopeta, Andhra Pradesh – 522601, www.nrtec.in**

DEPARTMENT OF

Artificial Intelligence & Machine Learning

CERTIFICATE

ACADEMIC YEAR: 2024-25

This is to certify that the bonafide record work done by______________________ ____________________________________bearing **Roll Number** ______________ of **II B.Tech. II Semester** in the **Full Stack Development-1 Laboratory** is satisfactorily completed.

Staff In-Charge **Head of the Department**

Submitted for the practical examination held on ______________________

External Examiner

INDEX

Week - 1 (Lists, Links and Images)

Aim :

Write a HTML program, to explain the working of lists. Note: It should have an ordered list, unordered list, nested lists and ordered list in an unordered list and definition lists.

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Lists Example</title>
</head>
<body>
    <h2>Ordered List</h2>
    <ol>
        <li>Item 1</li>
        <li>Item 2</li>
        <li>Item 3</li>
    </ol>

    <h2>Unordered List</h2>
    <ul>
        <li>Apple</li>
        <li>Banana</li>
        <li>Cherry</li>
    </ul>

    <h2>Nested List</h2>
    <ul>
        <li>Fruits
            <ul>
                <li>Apple</li>
```

```html
                <li>Banana</li>
            </ul>
        </li>
        <li>Vegetables
            <ul>
                <li>Carrot</li>
                <li>Broccoli</li>
            </ul>
        </li>
    </ul>

    <h2>Ordered List inside Unordered List</h2>
    <ul>
        <li>Steps to Make Tea:
            <ol>
                <li>Boil Water</li>
                <li>Add Tea Leaves</li>
                <li>Pour in Cup</li>
            </ol>
        </li>
    </ul>

    <h2>Definition List</h2>
    <dl>
        <dt>HTML</dt>
        <dd>HyperText Markup Language</dd>
        <dt>CSS</dt>
        <dd>Cascading Style Sheets</dd>
    </dl>
</body>
</html>
```

Output:

Ordered List

1. Item 1
2. Item 2
3. Item 3

Unordered List

- Apple
- Banana
- Cherry

Nested List

- Fruits
  - Apple
  - Banana
- Vegetables
  - Carrot
  - Broccoli

Ordered List inside Unordered List

- Steps to Make Tea:
  1. Boil Water
  2. Add Tea Leaves
  3. Pour in Cup

Definition List

HTML
: HyperText Markup Language

CSS
: Cascading Style Sheets

Aim :

Write a HTML program, to explain the working of hyperlinks using tag and href, target Attributes.

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Hyperlinks Example</title>
</head>
<body>
    <h2>Hyperlink Example</h2>
    <p>Click the link below to visit Google:</p>
    <a href="https://www.google.com" target="_blank">Visit
Google</a>
</body>
</html>
```

Output:

Hyperlink Example

Click the link below to visit Google:

Visit Google

Aim :

Create a HTML document that has your image and your friend's image with a specific height and width. Also when clicked on the images it should navigate to their respective profiles

File name: index.html

File Name :

index.html

Code :

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Image Links Example</title>
</head>
<body>
    <h2>Profile Images</h2>
    <p>Click on the images to visit the respective profiles:
</p>
    <a href="https://www.yourprofile.com" target="_blank">
        <img
src="https://upload.wikimedia.org/wikipedia/commons/9/9a/Gull_p
ortrait_ca_usa.jpg" alt="Your Image" width="150" height="150">
    </a>
    <a href="https://www.friendprofile.com" target="_blank">
        <img
src="https://upload.wikimedia.org/wikipedia/commons/6/6a/JavaSc
ript-logo.png" alt="Friend's Image" width="150" height="150">
    </a>
</body>
</html>
```

Output:

Profile Images

Click on the images to visit the respective profiles:

Aim :

Write a HTML program, in such a way that, rather than placing large images on a page, the preferred technique is to use thumbnails by setting the height and width parameters to something like to 100*100 pixels. Each thumbnail image is also a link to a full sized version of the image. Create an image gallery using this technique

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Image Gallery</title>
</head>
<body>
    <h2>Image Gallery</h2>
    <p>Click on the thumbnails to view full-sized images:</p>

    <a
href="https://upload.wikimedia.org/wikipedia/commons/9/9a/Gull_portrait_ca_usa.jpg" target="_blank">
        <img
src="https://upload.wikimedia.org/wikipedia/commons/9/9a/Gull_portrait_ca_usa.jpg" alt="Seagull" width="100" height="100">
    </a>
    <a
href="https://upload.wikimedia.org/wikipedia/commons/6/6a/JavaScript-logo.png" target="_blank">
        <img
src="https://upload.wikimedia.org/wikipedia/commons/6/6a/JavaScript-logo.png" alt="JavaScript Logo" width="100" height="100">
    </a>
```

```
    <a href="https://upload.wikimedia.org/wikipedia/commons/3/3a/Cat03.jpg" target="_blank">
        <img src="https://upload.wikimedia.org/wikipedia/commons/3/3a/Cat03.jpg" alt="Cat" width="100" height="100">
    </a>
</body>
</html>
```

Output:

Image Gallery

Click on the thumbnails to view full-sized images:

Week-2(HTML Tables, Forms and Frames)

Aim :

Write a HTML program, to explain the working of tables. (use tags: <table>,<tr>,<td>,attributes ,colspan,rowspan).

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>HTML Tables</title>
</head>
<body>
    <h2>Student Marks Table</h2>
    <table border="1">
        <tr>
            <th>Name</th>
            <th>Subject</th>
            <th colspan="2">Marks</th>
        </tr>
        <tr>
            <td rowspan="2">Alice</td>
            <td>Math</td>
            <td>90</td>
            <td>Excellent</td>
        </tr>
        <tr>
            <td>Science</td>
            <td>85</td>
            <td>Good</td>
        </tr>
        <tr>
```

```
            <td rowspan="2">Bob</td>
            <td>Math</td>
            <td>80</td>
            <td>Good</td>
        </tr>
        <tr>
            <td>Science</td>
            <td>88</td>
            <td>Very Good</td>
        </tr>
    </table>
</body>
</html>
```

Output:

Student Marks Table

Name	Subject	Marks	
Alice	Math	90	Excellent
	Science	85	Good
Bob	Math	80	Good
	Science	88	Very Good

Aim :

Write a HTML program, to explain the working of tables by preparing a timetable. (Note: Use tag to set the caption to the table & also use cell spacing, cell padding, border, rowspan, colspan etc.).

File name: index.html

File Name :

index.html

Code :

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Timetable</title>
</head>
<body>
    <h2>Class Timetable</h2>
    <table border="1" cellpadding="5" cellspacing="2">
        <caption>Weekly Class Schedule</caption>
        <tr>
            <th>Day</th>
            <th>9 AM - 10 AM</th>
            <th>10 AM - 11 AM</th>
            <th>11 AM - 12 PM</th>
            <th>12 PM - 1 PM</th>
            <th>Lunch Break</th>
            <th>2 PM - 3 PM</th>
            <th>3 PM - 4 PM</th>
        </tr>
        <tr>
            <td>Monday</td>
            <td>Math</td>
            <td>Science</td>
            <td>English</td>
```

```
                <td>History</td>
                <td rowspan="6">Break</td>
                <td>Physics</td>
                <td>Sports</td>
            </tr>
            <tr>
                <td>Tuesday</td>
                <td>Biology</td>
                <td>Chemistry</td>
                <td>Math</td>
                <td>English</td>
                <td>Computer Science</td>
                <td>Arts</td>
            </tr>
            <tr>
                <td>Wednesday</td>
                <td>Physics</td>
                <td>Math</td>
                <td>Science</td>
                <td>History</td>
                <td>Computer Science</td>
                <td>Sports</td>
            </tr>
            <tr>
                <td>Thursday</td>
                <td>Biology</td>
                <td>English</td>
                <td>Math</td>
                <td>Chemistry</td>
                <td>Physics</td>
                <td>Arts</td>
            </tr>
            <tr>
                <td>Friday</td>
                <td>English</td>
                <td>History</td>
                <td>Math</td>
                <td>Science</td>
                <td>Computer Science</td>
                <td>Sports</td>
            </tr>
            <tr>
                <td>Saturday</td>
```

```
            <td colspan="6">No Classes</td>
        </tr>
    </table>
</body>
</html>
```

Output:

Class Timetable

Weekly Class Schedule

Day	9 AM - 10 AM	10 AM - 11 AM	11 AM - 12 PM	12 PM - 1 PM	Lunch Break	2 PM - 3 PM	3 PM - 4 PM
Monday	Math	Science	English	History	Break	Physics	Sports
Tuesday	Biology	Chemistry	Math	English		Computer Science	Arts
Wednesday	Physics	Math	Science	History		Computer Science	Sports
Thursday	Biology	English	Math	Chemistry		Physics	Arts
Friday	English	History	Math	Science		Computer Science	Sports
Saturday	No Classes						

Aim :

Write a HTML program, to explain the working of forms by designing Registration form. (Note: Include text field, password field, number field, date of birth field, checkboxes, radio buttons, list boxes using and two buttons ie: submit and reset. Use tables to provide a better view).

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Registration Form</title>
</head>
<body>
    <h2>Registration Form</h2>
    <form>
        <table border="1" cellpadding="5" cellspacing="2">
            <tr>
                <td>Full Name:</td>
                <td><input type="text" name="fullname"
required></td>
            </tr>
            <tr>
                <td>Password:</td>
                <td><input type="password" name="password"
required></td>
            </tr>
            <tr>
                <td>Age:</td>
                <td><input type="number" name="age" required>
```

```
</td>
            </tr>
            <tr>
                <td>Date of Birth:</td>
                <td><input type="date" name="dob" required>
</td>
            </tr>
            <tr>
                <td>Gender:</td>
                <td>
                    <input type="radio" name="gender"
value="Male"> Male
                    <input type="radio" name="gender"
value="Female"> Female
                </td>
            </tr>
            <tr>
                <td>Hobbies:</td>
                <td>
                    <input type="checkbox" name="hobby"
value="Reading"> Reading
                    <input type="checkbox" name="hobby"
value="Sports"> Sports
                    <input type="checkbox" name="hobby"
value="Music"> Music
                </td>
            </tr>
            <tr>
                <td>Country:</td>
                <td>
                    <select name="country">
                        <option value="India">India</option>
                        <option value="USA">USA</option>
                        <option value="UK">UK</option>
                    </select>
                </td>
            </tr>
            <tr>
                <td colspan="2" align="center">
                    <button type="submit">Submit</button>
                    <button type="reset">Reset</button>
                </td>
            </tr>
```

```
            </table>
        </form>
    </body>
    </html>
```

Output:

Aim :

Write a HTML program, to explain the working of frames, such that page is to be divided into paragraph, third frameà image, second frame à3 parts on either direction. (Note: first frame)

File names: index.html, left.html, image.html, right.html, paragraph.html

File Name :

image.html

Code :

```
<!DOCTYPE html>
<html>
<head>
    <title>Image Frame</title>
</head>
<body>
    <img src="https://via.placeholder.com/300" alt="Placeholder
Image">
</body>
</html>
```

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Frames Example</title>
```

```
</head>
<frameset rows="30%, 70%">
    <frame src="paragraph.html" name="topFrame">
    <frameset cols="30%, 40%, 30%">
        <frame src="left.html" name="leftFrame">
        <frame src="image.html" name="middleFrame">
        <frame src="right.html" name="rightFrame">
    </frameset>
</frameset>
</html>
```

File Name :

left.html

Code :

```
<!DOCTYPE html>
<html>
<head>
    <title>Left Frame</title>
</head>
<body>
    <p>Left section of the second row.</p>
</body>
</html>
```

File Name :

paragraph.html

Code :

```
<!DOCTYPE html>
<html>
<head>
    <title>Paragraph Frame</title>
```

```html
</head>
<body>
    <p>This is the top frame containing a paragraph.</p>
</body>
</html>
```

File Name :

right.html

Code :

```html
<!DOCTYPE html>
<html>
<head>
    <title>Right Frame</title>
</head>
<body>
    <p>Right section of the second row.</p>
</body>
</html>
```

Output:

Week-3(HTML 5 and Cascading Style Sheets, Types of CSS)

Aim :

Write a HTML program, that makes use of <article> , <aside> ,<figher> ,<figcaption> <header>, <footer>, <main><nav><section><div><span> tags.

File name: index.html

File Name :

index.html

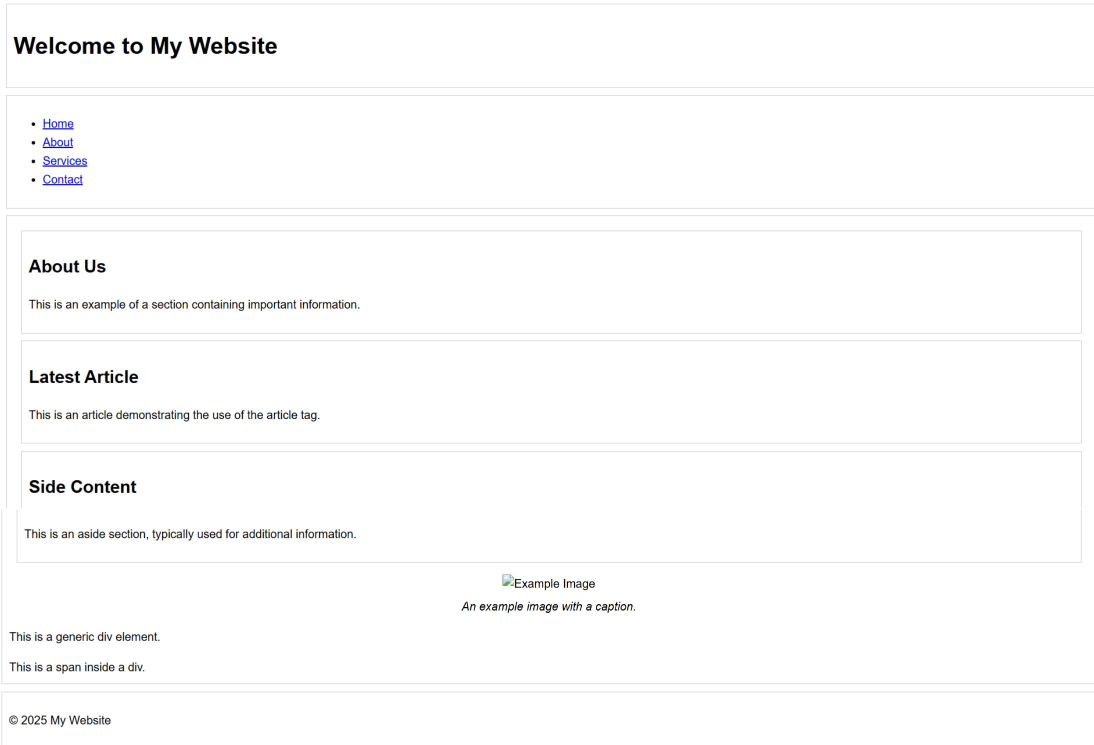
Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Semantic HTML Example</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 0;
            padding: 0;
            line-height: 1.6;
        }
        header, nav, main, section, article, aside, footer {
            padding: 10px;
            margin: 10px;
            border: 1px solid #ccc;
        }
        figure {
            display: flex;
            flex-direction: column;
            align-items: center;
        }
        figcaption {
```

```
            font-style: italic;
            margin-top: 5px;
        }
    </style>
</head>
<body>

    <header>
        <h1>Welcome to My Website</h1>
    </header>

    <nav>
        <ul>
            <li><a href="#">Home</a></li>
            <li><a href="#">About</a></li>
            <li><a href="#">Services</a></li>
            <li><a href="#">Contact</a></li>
        </ul>
    </nav>

    <main>
        <section>
            <h2>About Us</h2>
            <p>This is an example of a section containing
important information.</p>
        </section>

        <article>
            <h2>Latest Article</h2>
            <p>This is an article demonstrating the use of the
article tag.</p>
        </article>

        <aside>
            <h2>Side Content</h2>
            <p>This is an aside section, typically used for
additional information.</p>
        </aside>

        <figure>
            <img src="https://via.placeholder.com/300"
alt="Example Image">
            <figcaption>An example image with a caption.
```

```
</figcaption>
        </figure>

        <div>
            <p>This is a generic div element.</p>
            <span>This is a span inside a div.</span>
        </div>
    </main>

    <footer>
        <p>&copy; 2025 My Website</p>
    </footer>

</body>
</html>
```

Output:



Aim :

Write a HTML program, to embed audio and video into HTML web page.

Use index.html

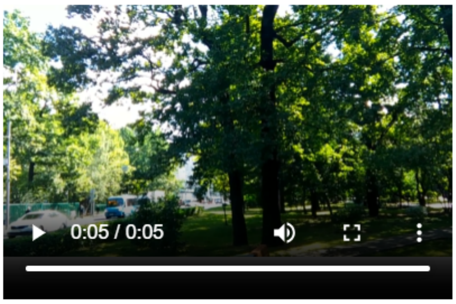
File Name :

index.html

Code :

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Embedded Audio and Video</title>
</head>
<body>
    <audio controls>
        <source
src="https://www.soundhelix.com/examples/mp3/SoundHelix-Song-
1.mp3" type="audio/mpeg">
        Your browser does not support the audio element.
    </audio>
    <br>
    <video width="320" height="240" controls>
        <source
src="https://samplelib.com/lib/preview/mp4/sample-5s.mp4"
type="video/mp4">
        Your browser does not support the video tag.
    </video>
</body>
</html>
```

Output:



Aim :

Write a program to apply different types (or levels of styles or style specification formats) - inline, internal, external styles to HTML elements. (identify selector, property and value)
File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>CSS Styling Levels Example</title>
    <link rel="stylesheet" href="styles.css">  <!-- External
CSS -->
    <style>
        /* Internal CSS */
        h1 {
            color: darkblue;
            text-align: center;
        }
        .styled-div {
            background-color: lightgray;
            padding: 20px;
            border: 1px solid black;
        }
    </style>
</head>
<body>


    <h1>Applying Different CSS Styles</h1>
```

```html
    <!-- Inline CSS -->
    <p style="color: red; font-size: 18px;">This paragraph uses inline styling.</p>


    <!-- Internal CSS Applied -->
    <div class="styled-div">
        <p>This div is styled using internal CSS.</p>
    </div>


    <!-- External CSS Applied -->
    <p class="external-style">This paragraph is styled using external CSS.</p>


</body>
</html>
```

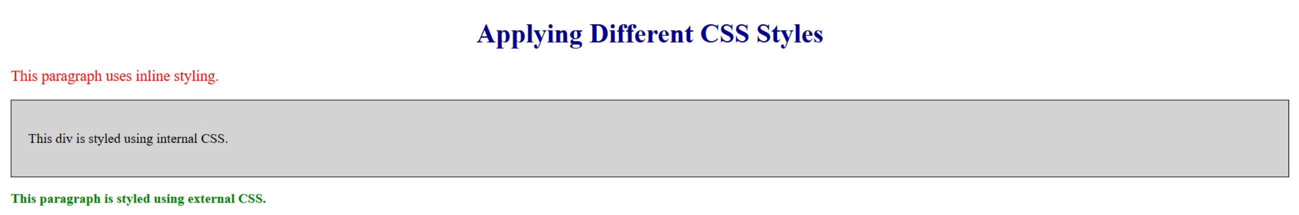
File Name :

styles.css

Code :

```css
.external-style {
    color: green;
    font-weight: bold;
}
```

Output:



Week-4(Selector forms)

Aim :

Write a program to apply different types of selector forms

1. Simple selector (element, id, class, group, universal)
2. Combinator selector (descendant, child, adjacent sibling, general sibling)
3. Pseudo-class selector
4. Pseudo-element selector
5. Attribute selector

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>CSS Selectors</title>
    <link rel="stylesheet" href="styles.css"> <!-- External CSS
-->
</head>
<body>
    <h1 id="main-title">CSS Selectors Example</h1>
    <p class="highlight">This is a paragraph with a class
selector.</p>
    <p>This is another paragraph.</p>
    <div>
        <span class="highlight">This is inside a div and has a
class.</span>
    </div>
    <ul>
        <li>First Item</li>
        <li>Second Item</li>
        <li>Third Item</li>
    </ul>
```

```
    <a href="#" title="Hover over me">Hover over me</a>
</body>
</html>
```

File Name :

styles.css

Code :

```
h1 {
    color: blue;
}

#main-title {
    text-align: center;
}

.highlight {
    font-weight: bold;
    color: red;
}

/* Combinator Selectors */
div span {
    background-color: yellow;
}

ul > li {
    list-style-type: square;
}

/* Pseudo-Class Selector */
a:hover {
    color: green;
}

/* Pseudo-Element Selector */
p::first-letter {
    font-size: 2em;
}
```

```
/* Attribute Selector */
a[title] {
    text-decoration: underline;
}
```

Output:

Week-5(CSS with Color, Background, Font, Text and CSS Box Model)

Aim :

Write a program to demonstrate the various ways you can reference a color in CSS.

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>CSS Colors Example</title>
    </head>
<body>
<style>
.named-color {
    color: red;
}

/* HEX Color */
.hex-color {
    color: #ff5733;
}

/* RGB Color */
.rgb-color {
    color: rgb(34, 139, 34);
}

/* HSL Color */
.hsl-color {
    color: hsl(240, 100%, 50%);
```

```
    }

    /* RGBA Color with transparency */
    .rgba-color {
        color: rgba(255, 99, 71, 0.5);
    }

    </style>
        <h1 id="main-title">CSS Colors Example</h1>
        <p class="named-color">This text uses a named color.</p>
        <p class="hex-color">This text uses a HEX color.</p>
        <p class="rgb-color">This text uses an RGB color.</p>
        <p class="hsl-color">This text uses an HSL color.</p>
        <p class="rgba-color">This text uses an RGBA color.</p>
    </body>
    </html>
```

Output:

CSS Colors Example

This text uses a named color.

This text uses a HEX color.

This text uses an RGB color.

This text uses an HSL color.

This text uses an RGBA color.

Aim :

Write a CSS rule that places a background image halfway down the page, tilting it horizontally. The image should remain in place when the user scrolls up or down.

File name: index.html

File Name :

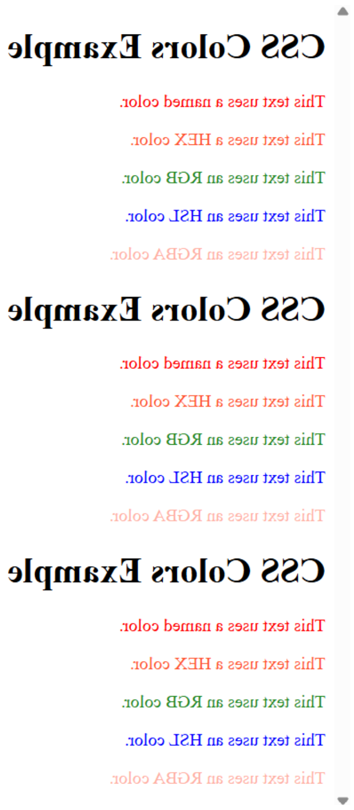
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>CSS Colors Example</title>
    <style>
        /* Named Color */
        .named-color {
            color: red;
        }

        /* HEX Color */
        .hex-color {
            color: #ff5733;
        }

        /* RGB Color */
        .rgb-color {
            color: rgb(34, 139, 34);
        }

        /* HSL Color */
        .hsl-color {
            color: hsl(240, 100%, 50%);
        }
```

```
        /* RGBA Color with transparency */
        .rgba-color {
            color: rgba(255, 99, 71, 0.5);
        }

        /* Background Image Styling */
        body {
            background-image:
url('https://via.placeholder.com/1500');
            background-repeat: no-repeat;
            background-position: center bottom;
            background-size: 100% 50%;
            background-attachment: fixed;
            transform: scaleX(-1);
            overflow-y: scroll;
        }
    </style>
</head>
<body>
    <h1 id="main-title">CSS Colors Example</h1>
    <p class="named-color">This text uses a named color.</p>
    <p class="hex-color">This text uses a HEX color.</p>
    <p class="rgb-color">This text uses an RGB color.</p>
    <p class="hsl-color">This text uses an HSL color.</p>
    <p class="rgba-color">This text uses an RGBA color.</p>
<h1 id="main-title">CSS Colors Example</h1>
    <p class="named-color">This text uses a named color.</p>
    <p class="hex-color">This text uses a HEX color.</p>
    <p class="rgb-color">This text uses an RGB color.</p>
    <p class="hsl-color">This text uses an HSL color.</p>
    <p class="rgba-color">This text uses an RGBA color.</p>

<h1 id="main-title">CSS Colors Example</h1>
    <p class="named-color">This text uses a named color.</p>
    <p class="hex-color">This text uses a HEX color.</p>
    <p class="rgb-color">This text uses an RGB color.</p>
    <p class="hsl-color">This text uses an HSL color.</p>
    <p class="rgba-color">This text uses an RGBA color.</p>

</body>
</html>
```

Output:



CSS Colors Example
This text uses a named color.
This text uses a HEX color.
This text uses an RGB color.
This text uses an HSL color.
This text uses an RGBA color.
CSS Colors Example
This text uses a named color.
This text uses a HEX color.
This text uses an RGB color.
This text uses an HSL color.
This text uses an RGBA color.
CSS Colors Example
This text uses a named color.
This text uses a HEX color.
This text uses an RGB color.
This text uses an HSL color.
This text uses an RGBA color.

Aim :

Write a program using the following terms related to CSS font and text:

i. font-size

ii. font-weight

iii. font-style

iv. text-decoration

v. text-transformation

vi. text-alignment

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>CSS Font and Text Example</title>
    <style>
        /* Font and Text Styling */
        .font-size {
            font-size: 20px;
        }

        .font-weight {
            font-weight: bold;
        }

        .font-style {
            font-style: italic;
        }

        .text-decoration {
```

```
            text-decoration: underline;
        }

        .text-transform {
            text-transform: uppercase;
        }

        .text-align {
            text-align: center;
        }
    </style>
</head>
<body>
    <h1 id="main-title">CSS Font and Text Example</h1>
    <p class="font-size">This text has a font size of 20px.</p>
    <p class="font-weight">This text is bold.</p>
    <p class="font-style">This text is italic.</p>
    <p class="text-decoration">This text is underlined.</p>
    <p class="text-transform">This text is uppercase.</p>
    <p class="text-align">This text is center-aligned.</p>
</body>
</html>
```

Output:

CSS Font and Text Example

This text has a font size of 20px.

This text is bold.

This text is italic.

This text is underlined.

THIS TEXT IS UPPERCASE.

This text is center-aligned.

Aim :

Write a program, to explain the importance of CSS Box model using

i. Content ii. Border iii. Margin iv. padding

File name: index.html

File Name :

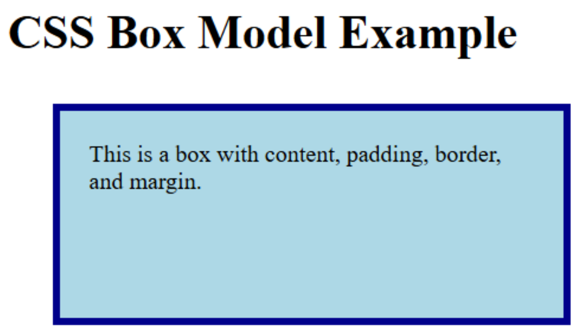
index.html

Code :

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>CSS Box Model Example</title>
    <style>
        /* Box Model Styling */
        .box-model {
            width: 300px;
            height: 100px;
            background-color: lightblue;
            padding: 20px;
            border: 5px solid darkblue;
            margin: 30px;
        }
    </style>
</head>
<body>
    <h1 id="main-title">CSS Box Model Example</h1>
    <div class="box-model">This is a box with content, padding,
border, and margin.</div>
</body>
```

```
</html>
```

Output:


CSS Box Model Example
This is a box with content, padding, border, and margin.

Week-6(Applying JavaScript - internal and external, I/O, Type Conversion)

Aim :

Write a program to embed internal and external JavaScript in a web page.

File names: index.html , script.js

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>JavaScript Example</title>
    <script>
        // Internal JavaScript
        function showAlert() {
            alert('This is an internal JavaScript function!');
        }
    </script>
    <script src="script.js"></script> <!-- External JavaScript
-->
</head>
<body>
    <h1 id="main-title">JavaScript Example</h1>
    <button onclick="showAlert()">Click for Internal
JS</button>
    <button onclick="showExternalAlert()">Click for External
JS</button>
</body>
</html>
```

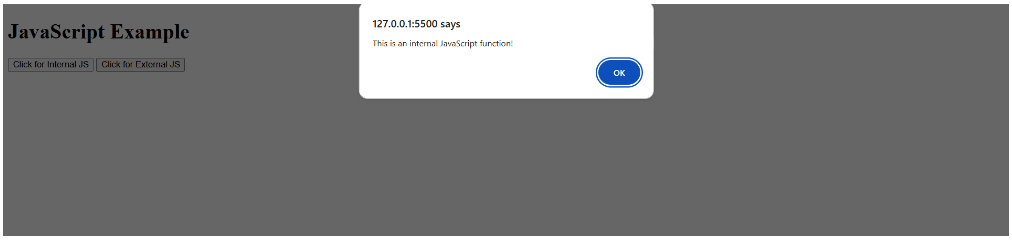
File Name :

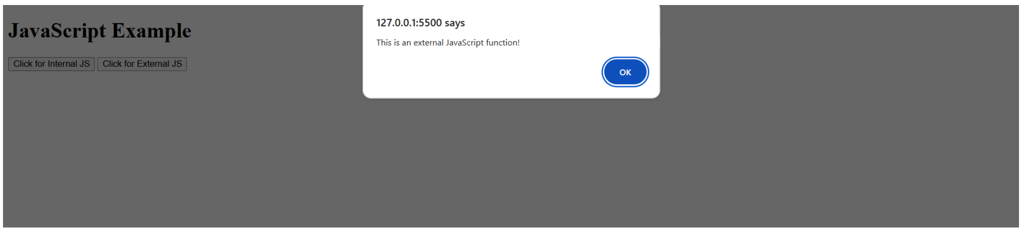
script.js

Code :

```
function showExternalAlert() {
    alert('This is an external JavaScript function!');
}
```

Output:





Aim :

Write a program to explain the different ways for displaying output.

File names: index.html, script.js

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>JavaScript Output Methods</title>
    <script>
        // Internal JavaScript
        function showAlert() {
            alert('This is an alert box!');
        }

        function writeToDocument() {
            document.write('This text is written using
document.write().');
        }

        function logToConsole() {
            console.log('This message is logged to the
console.');
        }

        function changeInnerHTML() {
            document.getElementById('output').innerHTML = 'This
text is set using innerHTML!';
        }
    </script>
    <script src="script.js"></script> <!-- External JavaScript
```

```
    -->
    </head>
    <body>
        <h1 id="main-title">JavaScript Output Methods</h1>
        <button onclick="showAlert()">Show Alert</button>
        <button onclick="writeToDocument()">Write to
    Document</button>
        <button onclick="logToConsole()">Log to Console</button>
        <button onclick="changeInnerHTML()">Change Inner
    HTML</button>
        <p id="output"></p>
    </body>
    </html>
```

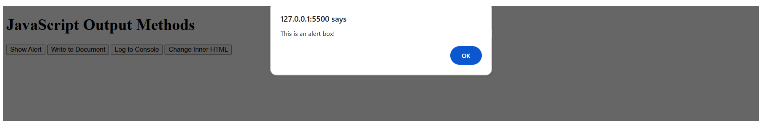
File Name :

script.js

Code :

```
function showExternalAlert() {
    alert('This is an external JavaScript function!');
}
```

Output:



This text is written using document.write().

Aim :

Write a program to explain the different ways for taking input.

File name: index.html, script.js

File Name :

index.html

Code :

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>JavaScript Input Methods</title>
    <script>
        // Internal JavaScript
        function getPromptInput() {
            let userInput = prompt('Enter something:');
            document.getElementById('output').innerHTML =
'Prompt Input: ' + userInput;
        }

        function getFormInput() {
            let formInput =
document.getElementById('textInput').value;
            document.getElementById('output').innerHTML = 'Form
Input: ' + formInput;
        }
    </script>
    <script src="script.js"></script> <!-- External JavaScript
-->
</head>
<body>
    <h1 id="main-title">JavaScript Input Methods</h1>
    <button onclick="getPromptInput()">Use Prompt</button>
```

```
    <input type="text" id="textInput" placeholder="Type
something">
    <button onclick="getFormInput()">Get Form Input</button>
    <p id="output"></p>
</body>
</html>
```

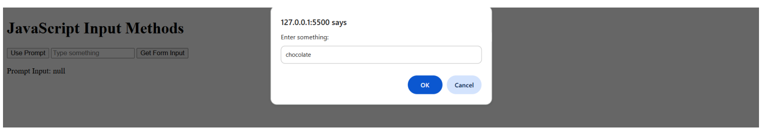
File Name :

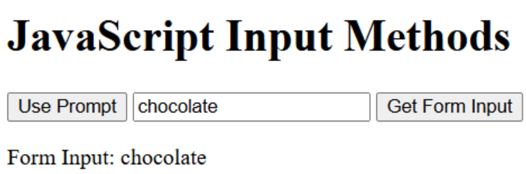
script.js

Code :

```
function getExternalInput() {
    let userInput = prompt('External Script Prompt: Enter
something');
    document.getElementById('output').innerHTML = 'External
Script Input: ' + userInput;
}
```

Output:





Aim :

Create a webpage which uses prompt dialogue box to ask a voter for his name and age. Display the information in table format along with either the voter can vote or not

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Voter Eligibility Check</title>
    <script>
        function checkVoterEligibility() {
            let name = prompt('Enter your name:');
            let age = prompt('Enter your age:');
            let eligibility = (age >= 18) ? 'Eligible to Vote'
: 'Not Eligible to Vote';

            document.getElementById('voterTable').innerHTML = `
                <tr>
                    <th>Name</th>
                    <th>Age</th>
                    <th>Eligibility</th>
                </tr>
                <tr>
                    <td>${name}</td>
                    <td>${age}</td>
                    <td>${eligibility}</td>
                </tr>
            `;
        }
```

```
    </script>
</head>
<body onload="checkVoterEligibility()">
    <h1>Voter Eligibility Check</h1>
    <table border="1" id="voterTable"></table>
</body>
</html>
```

Output:

Voter Eligibility Check

Name	Age	Eligibility
null	null	Not Eligible to Vote

Week-7(Java Script Pre-defined and User-defined Objects)

Aim :

Write a program using document object properties and methods.

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Original Title</title>
    <script>
        document.addEventListener("DOMContentLoaded", function
() {
            // 1. Update the document title
            document.title = "Updated Document Title";

            // 2. Modify existing heading content
            const header = document.getElementById("header");
            if (header) {
                header.innerHTML = "Updated Heading";
            }

            // 3. Add a new paragraph dynamically
            const newParagraph = document.createElement("p");
            newParagraph.textContent = "This is a dynamically
added paragraph.";
            document.body.appendChild(newParagraph);
        });
    </script>
</head>
```

```html
<body>

    <h1 id="header">Original Heading</h1>

</body>
</html>
```

Output:

Updated Heading

This is a dynamically added paragraph.

Aim :

Write a program using window object properties and methods.

File name: index.html

File Name :

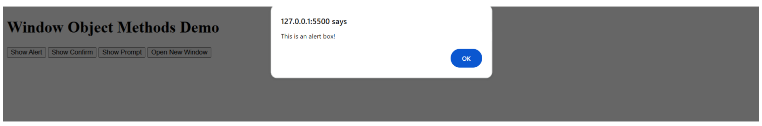
index.html

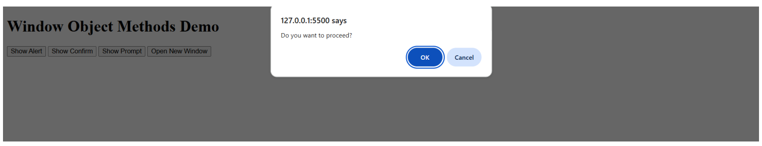
Code :

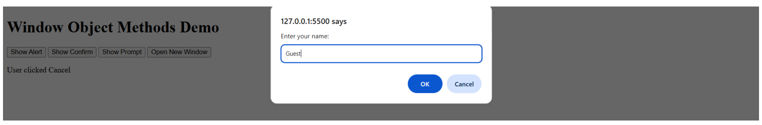
```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Window Object Demo</title>
    <script>
        function showAlert() {
            window.alert("This is an alert box!");
        }

        function showConfirm() {
            let result = window.confirm("Do you want to
proceed?");

document.getElementById("confirmResult").textContent = result ?
"User clicked OK" : "User clicked Cancel";
        }

        function showPrompt() {
            let name = window.prompt("Enter your name:",
"Guest");
            document.getElementById("promptResult").textContent
= name ? "Hello, " + name : "No name entered";
        }

        function openNewWindow() {
            window.open("https://www.example.com", "_blank",
```

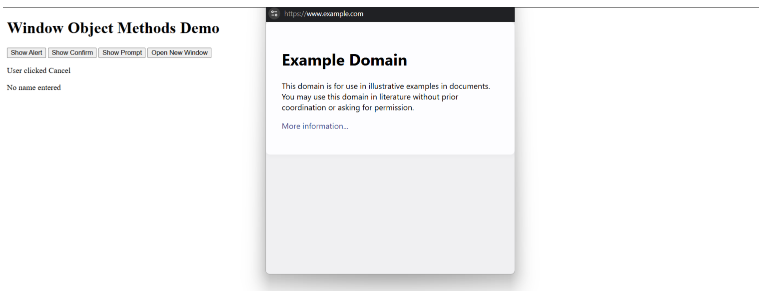
```
"width=500,height=500");
        }
    </script>
</head>
<body>
    <h1>Window Object Methods Demo</h1>
    <button onclick="showAlert()">Show Alert</button>
    <button onclick="showConfirm()">Show Confirm</button>
    <button onclick="showPrompt()">Show Prompt</button>
    <button onclick="openNewWindow()">Open New Window</button>

    <p id="confirmResult"></p>
    <p id="promptResult"></p>
</body>
</html>
```

Output:









Aim :

Write a program using array object properties and methods.

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Array Object Demo</title>
    <script>
        function showArrayMethods() {
            let numbers = [1, 2, 3, 4, 5];

document.getElementById("originalArray").textContent =
"Original Array: " + numbers;

            numbers.push(6);
            document.getElementById("pushResult").textContent =
"After Push (6 added): " + numbers;

            numbers.pop();
            document.getElementById("popResult").textContent =
"After Pop (last element removed): " + numbers;

            numbers.shift();
            document.getElementById("shiftResult").textContent
= "After Shift (first element removed): " + numbers;

            numbers.unshift(0);

document.getElementById("unshiftResult").textContent = "After
```

```
Unshift (0 added at start): " + numbers;

            let joined = numbers.join(" - ");
            document.getElementById("joinResult").textContent =
"Joined Array: " + joined;
        }
    </script>
</head>
<body>
    <h1>Array Object Methods Demo</h1>
    <button onclick="showArrayMethods()">Show Array
Methods</button>
    <p id="originalArray"></p>
    <p id="pushResult"></p>
    <p id="popResult"></p>
    <p id="shiftResult"></p>
    <p id="unshiftResult"></p>
    <p id="joinResult"></p>
</body>
</html>
```

Output:

Array Object Methods Demo

Show Array Methods

Original Array: 1,2,3,4,5

After Push (6 added): 1,2,3,4,5,6

After Pop (last element removed): 1,2,3,4,5

After Shift (first element removed): 2,3,4,5

After Unshift (0 added at start): 0,2,3,4,5

Joined Array: 0 - 2 - 3 - 4 - 5

Aim :

Write a program using math object properties and methods.

File name: index.html

File Name :

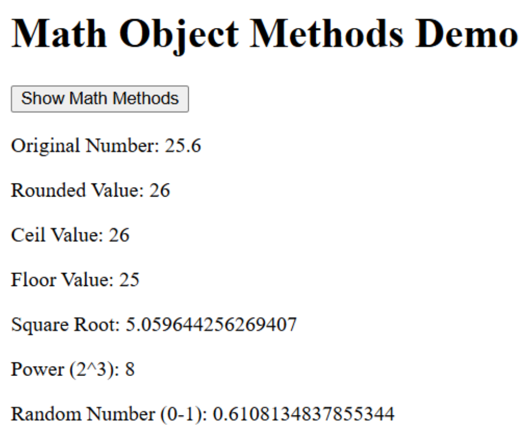
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Math Object Demo</title>
    <script>
        function showMathMethods() {
            let num = 25.6;

document.getElementById("originalNumber").textContent =
"Original Number: " + num;

            document.getElementById("roundResult").textContent
= "Rounded Value: " + Math.round(num);
            document.getElementById("ceilResult").textContent =
"Ceil Value: " + Math.ceil(num);
            document.getElementById("floorResult").textContent
= "Floor Value: " + Math.floor(num);
            document.getElementById("sqrtResult").textContent =
"Square Root: " + Math.sqrt(num);
            document.getElementById("powResult").textContent =
"Power (2^3): " + Math.pow(2, 3);
            document.getElementById("randomResult").textContent
= "Random Number (0-1): " + Math.random();
        }
    </script>
</head>
```

```
<body>
    <h1>Math Object Methods Demo</h1>
    <button onclick="showMathMethods()">Show Math
Methods</button>
    <p id="originalNumber"></p>
    <p id="roundResult"></p>
    <p id="ceilResult"></p>
    <p id="floorResult"></p>
    <p id="sqrtResult"></p>
    <p id="powResult"></p>
    <p id="randomResult"></p>
</body>
</html>
```

Output:



Aim :

Write a program using string object properties and methods.

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>String Object Demo</title>
    <script>
        function showStringMethods() {
            let text = " Hello, JavaScript! ";

document.getElementById("originalString").textContent =
"Original String: " + text;

            document.getElementById("lengthResult").textContent
= "String Length: " + text.length;
            document.getElementById("trimResult").textContent =
"Trimmed String: '" + text.trim() + "'";

document.getElementById("uppercaseResult").textContent =
"Uppercase: " + text.toUpperCase();

document.getElementById("lowercaseResult").textContent =
"Lowercase: " + text.toLowerCase();

document.getElementById("substringResult").textContent =
"Substring (0,5): " + text.substring(0, 5);

document.getElementById("replaceResult").textContent = "Replace
```

```
'JavaScript' with 'World': " + text.replace("JavaScript",
"World");
        }
    </script>
</head>
<body>
    <h1>String Object Methods Demo</h1>
    <button onclick="showStringMethods()">Show String
Methods</button>
    <p id="originalString"></p>
    <p id="lengthResult"></p>
    <p id="trimResult"></p>
    <p id="uppercaseResult"></p>
    <p id="lowercaseResult"></p>
    <p id="substringResult"></p>
    <p id="replaceResult"></p>
</body>
</html>
```

Output:

Aim :

Write a program using regex object properties and methods.

File name: index.html

File Name :

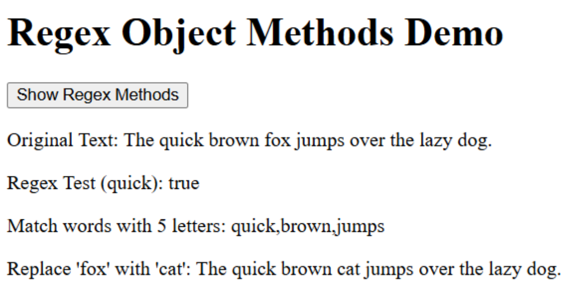
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Regex Object Demo</title>
    <script>
        function showRegexMethods() {
            let text = "The quick brown fox jumps over the lazy
dog.";
            document.getElementById("originalText").textContent
= "Original Text: " + text;

            let regex = /quick/;
            document.getElementById("testResult").textContent =
"Regex Test (quick): " + regex.test(text);

            let match = text.match(/\b\w{5}\b/g);
            document.getElementById("matchResult").textContent
= "Match words with 5 letters: " + match;

            let replacedText = text.replace(/fox/, "cat");

document.getElementById("replaceResult").textContent = "Replace
'fox' with 'cat': " + replacedText;
        }
    </script>
</head>
```

```html
<body>
    <h1>Regex Object Methods Demo</h1>
    <button onclick="showRegexMethods()">Show Regex Methods</button>
    <p id="originalText"></p>
    <p id="testResult"></p>
    <p id="matchResult"></p>
    <p id="replaceResult"></p>
</body>
</html>
```

Output:



Aim :

Write a program using date object properties and methods.

File name: index.html

File Name :

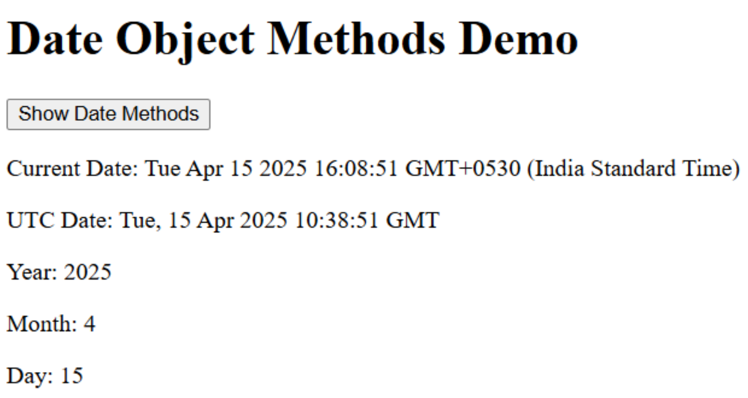
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Date Object Demo</title>
    <script>
        function showDateMethods() {
            let currentDate = new Date();
            document.getElementById("currentDate").textContent
= "Current Date: " + currentDate;

            document.getElementById("utcDate").textContent =
"UTC Date: " + currentDate.toUTCString();

            document.getElementById("year").textContent =
"Year: " + currentDate.getFullYear();

            document.getElementById("month").textContent =
"Month: " + (currentDate.getMonth() + 1);

            document.getElementById("day").textContent = "Day:
" + currentDate.getDate();
        }
    </script>
</head>
<body>
```

```
    <h1>Date Object Methods Demo</h1>
    <button onclick="showDateMethods()">Show Date
Methods</button>
    <p id="currentDate"></p>
    <p id="utcDate"></p>
    <p id="year"></p>
    <p id="month"></p>
    <p id="day"></p>
</body>
</html>
```

Output:



Aim :

Write a program to explain user-defined object by using properties, methods, accessors, constructors and display

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>User-Defined Object Demo</title>
    <script>
        class Person {
            constructor(name, age) {
                this._name = name;
                this._age = age;
            }

            // Getter for name
            get name() {
                return this._name;
            }

            // Setter for name
            set name(newName) {
                this._name = newName;
            }

            // Getter for age
            get age() {
                return this._age;
            }
```

```
            // Method to display details
            displayInfo() {
                return `Name: ${this.name}, Age: ${this.age}`;
            }
        }

        function showPersonDetails() {
            let person = new Person("John Doe", 30);
            document.getElementById("personInfo").textContent =
person.displayInfo();
        }
    </script>
</head>
<body>
    <h1>User-Defined Object Demo</h1>
    <button onclick="showPersonDetails()">Show Person
Details</button>
    <p id="personInfo"></p>
</body>
</html>
```

Output:

Week-8(Java Script Conditional Statements and Loops)

Aim :

Write a program which asks the user to enter three integers, obtains the numbers from the user and outputs HTML text that displays the larger number followed by the words “LARGER NUMBER” in an information message dialog. If the numbers are equal, output HTML text as “EQUAL NUMBERS”.

File name: index.html

File Name :

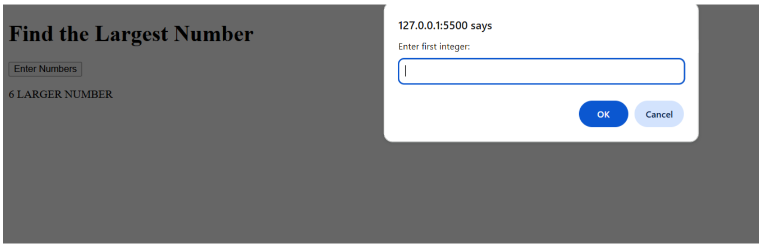
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Find Largest Number</title>
    <script>
        function findLargestNumber() {
            let num1 = parseInt(prompt("Enter first integer:"),
10);
            let num2 = parseInt(prompt("Enter second
integer:"), 10);
            let num3 = parseInt(prompt("Enter third integer:"),
10);

            let message;
            if (num1 === num2 && num2 === num3) {
                message = "EQUAL NUMBERS";
            } else {
                let largest = Math.max(num1, num2, num3);
                message = `${largest} LARGER NUMBER`;
            }
            document.getElementById("result").textContent =
message;
```

```
        }
    </script>
</head>
<body>
    <h1>Find the Largest Number</h1>
    <button onclick="findLargestNumber()">Enter
Numbers</button>
    <p id="result"></p>
</body>
</html>
```

Output:



Aim :

Write a program to display week days using switch case.

File name: index.html

File Name :

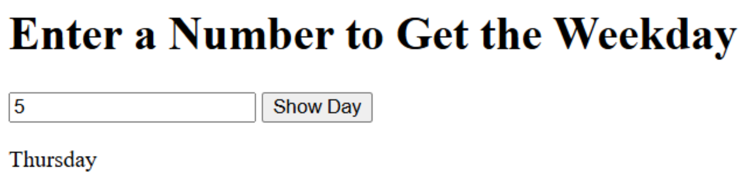
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Weekday Finder</title>
    <script>
        function getWeekday() {
            let dayNumber =
parseInt(document.getElementById("dayInput").value, 10);
            let dayName;

            switch (dayNumber) {
                case 1: dayName = "Sunday"; break;
                case 2: dayName = "Monday"; break;
                case 3: dayName = "Tuesday"; break;
                case 4: dayName = "Wednesday"; break;
                case 5: dayName = "Thursday"; break;
                case 6: dayName = "Friday"; break;
                case 7: dayName = "Saturday"; break;
                default: dayName = "Invalid input. Enter a
number between 1 and 7.";
            }

            document.getElementById("result").textContent =
dayName;
        }
    </script>
```

```
</head>
<body>
    <h1>Enter a Number to Get the Weekday</h1>
    <input type="number" id="dayInput" placeholder="Enter a
number (1-7)">
    <button onclick="getWeekday()">Show Day</button>
    <p id="result"></p>
</body>
</html>
```

Output:



Aim :

Write a program to print 1 to 10 numbers using for, while and do-while loops.

File name: index.html

File Name :

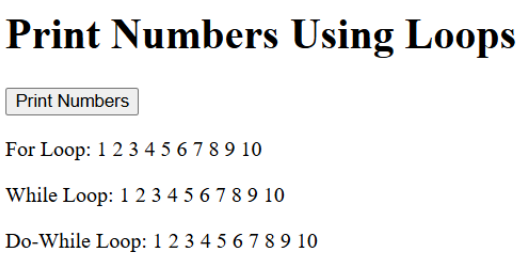
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Print Numbers Using Loops</title>
    <script>
        function printNumbers() {
            let forLoopResult = "For Loop: ";
            for (let i = 1; i <= 10; i++) {
                forLoopResult += i + " ";
            }

            let whileLoopResult = "While Loop: ";
            let j = 1;
            while (j <= 10) {
                whileLoopResult += j + " ";
                j++;
            }

            let doWhileLoopResult = "Do-While Loop: ";
            let k = 1;
            do {
                doWhileLoopResult += k + " ";
                k++;
            } while (k <= 10);

            document.getElementById("result").innerHTML =
```

```
                    `<p>${forLoopResult}</p>
                    <p>${whileLoopResult}</p>
                    <p>${doWhileLoopResult}</p>`;
            }
        </script>
    </head>
    <body>
        <h1>Print Numbers Using Loops</h1>
        <button onclick="printNumbers()">Print Numbers</button>
        <div id="result"></div>
    </body>
    </html>
```

Output:



Aim :

Write a program to print data in object using for-in, for-each and for-of loops.

File name: index.html

File Name :

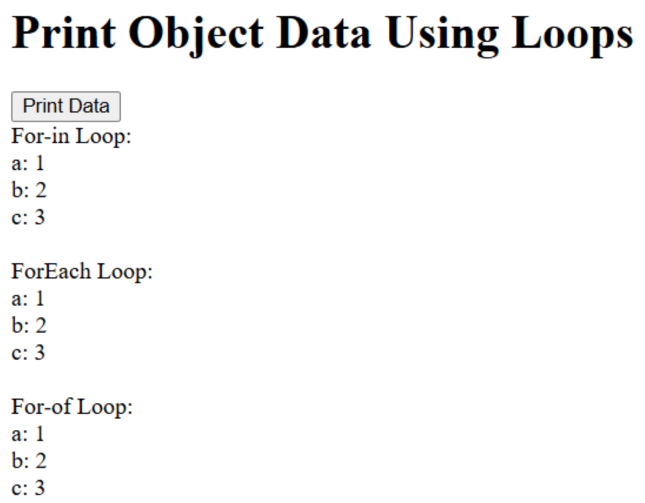
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Print Object Data</title>
    <script>
        function printObjectData() {
            const obj = { a: 1, b: 2, c: 3 };
            const output = [];

            // Using for-in loop
            let forInResult = "For-in Loop:<br>";
            for (let key in obj) {
                forInResult += `${key}: ${obj[key]}<br>`;
                output.push(`${key}: ${obj[key]}`);
            }

            // Using forEach (on Object.entries)
            let forEachResult = "ForEach Loop:<br>";
            Object.entries(obj).forEach(([key, value]) => {
                forEachResult += `${key}: ${value}<br>`;
                output.push(`${key}: ${value}`);
            });

            // Using for-of (on Object.entries)
            let forOfResult = "For-of Loop:<br>";
            for (let [key, value] of Object.entries(obj)) {
```

```
                forOfResult += `${key}: ${value}<br>`;
                output.push(`${key}: ${value}`);
            }

            // Display results
            document.getElementById("result").innerHTML =
forInResult + "<br>" + forEachResult + "<br>" + forOfResult;
        }
    </script>
</head>
<body>
    <h1>Print Object Data Using Loops</h1>
    <button onclick="printObjectData()">Print Data</button>
    <div id="result"></div>
</body>
</html>
```

Output:



Aim :

Develop a program to determine whether a given number is an ‘ARMSTRONG NUMBER’ or not. [Eg: 153 is an Armstrong number, since sum of the cube of the digits is equal to the number i.e.,13 + 53+ 33 = 153]

File name: index.html

File Name :

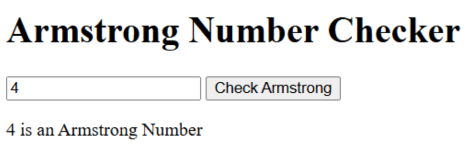
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Armstrong Number Checker</title>
    <script>
        function isArmstrongNumber(num) {
            let sum = 0;
            let digits = num.toString().split('').map(Number);
            let power = digits.length;

            digits.forEach(digit => {
                sum += Math.pow(digit, power);
            });

            return sum === num;
        }

        function checkArmstrong() {
            const input =
document.getElementById("numberInput").value.trim();
            const result = document.getElementById("result");

            if (!/^\d+$/.test(input)) {
                result.textContent = "Please enter a valid
```

```
number.";
                return;
            }

            const num = parseInt(input, 10);
            if (isArmstrongNumber(num)) {
                result.textContent = `${num} is an Armstrong
Number`;
            } else {
                result.textContent = `${num} is not an
Armstrong Number`;
            }
        }
    </script>
</head>
<body>
    <h1>Armstrong Number Checker</h1>
    <input type="text" id="numberInput" placeholder="Enter a
number">
    <button onclick="checkArmstrong()">Check Armstrong</button>
    <p id="result"></p>
</body>
</html>
```

Output:



Aim :

Write a program to display the denomination of the amount deposited in the bank in terms of 100's, 50's, 20's, 10's, 5's, 2's & 1's. (Eg: If deposited amount is Rs.163, the output should be 1-100's, 1-50's, 1- 10's, 1-2's & 1-1's)

File name: index.html

File Name :

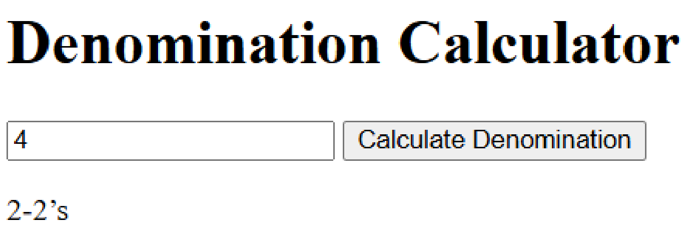
index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Denomination Calculator</title>
    <script>
        function calculateDenomination() {
            const amountInput =
document.getElementById("amount").value;
            const resultDiv =
document.getElementById("result");
            let amount = parseInt(amountInput);

            if (isNaN(amount) || amount <= 0) {
                resultDiv.textContent = "Please enter a valid
amount.";
                return;
            }

            const denominations = [100, 50, 20, 10, 5, 2, 1];
            let output = [];

            denominations.forEach(denomination => {
                if (amount >= denomination) {
                    let count = Math.floor(amount /
```

```
denomination);
                    output.push(`${count}-${denomination}’s`);
                    amount %= denomination;
                }
            });

            resultDiv.textContent = output.join(", ");
        }
    </script>
</head>
<body>

    <h1>Denomination Calculator</h1>
    <input type="text" id="amount" placeholder="Enter amount">
    <button onclick="calculateDenomination()">Calculate
Denomination</button>
    <p id="result"></p>

</body>
</html>
```

Output:



Week-9(: Java Script Functions and Events)

Aim :

Design a appropriate function should be called to display

1. Factorial of that number
2. Fibonacci series up to that number
3. Prime numbers up to that number
4. Is it palindrome or not

File name : index.html

File Name :

index.html

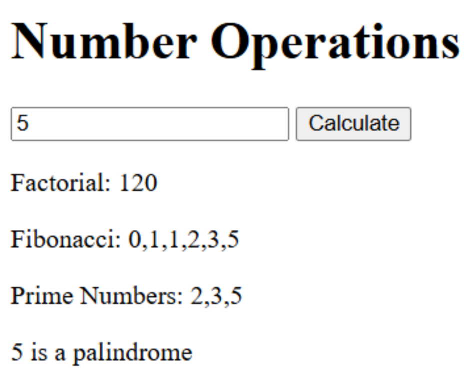
Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Number Operations</title>
    <script>
        function factorial(n) {
            if (n < 0) return "Factorial not defined for
negative numbers";
            if (n === 0 || n === 1) return 1;
            let result = 1;
            for (let i = 2; i <= n; i++) {
                result *= i;
            }
            return result;
        }

        function fibonacci(n) {
            if (n < 1) return "Enter a positive integer";
            let fibSeries = [0, 1];
            while (true) {
                let next = fibSeries[fibSeries.length - 1] +
fibSeries[fibSeries.length - 2];
```

```
                if (next > n) break;
                fibSeries.push(next);
            }
            return fibSeries;
        }

        function primesUpTo(n) {
            if (n < 2) return "No prime numbers";
            let primes = [];
            for (let i = 2; i <= n; i++) {
                let isPrime = true;
                for (let j = 2; j * j <= i; j++) {
                    if (i % j === 0) {
                        isPrime = false;
                        break;
                    }
                }
                if (isPrime) primes.push(i);
            }
            return primes;
        }

        function isPalindrome(n) {
            let str = n.toString();
            let reversedStr = str.split('').reverse().join('');
            return str === reversedStr ? `${n} is a palindrome`
: `${n} is not a palindrome`;
        }

        function displayResults() {
            let num =
parseInt(document.getElementById("numberInput").value, 10);

document.getElementById("factorialResult").textContent =
`Factorial: ${factorial(num)}`;

document.getElementById("fibonacciResult").textContent =
`Fibonacci: ${fibonacci(num)}`;
            document.getElementById("primeResult").textContent
= `Prime Numbers: ${primesUpTo(num)}`;

document.getElementById("palindromeResult").textContent =
isPalindrome(num);
```

```
        }
    </script>
</head>
<body>
    <h1>Number Operations</h1>
    <input type="number" id="numberInput" placeholder="Enter a
number">
    <button onclick="displayResults()">Calculate</button>
    <p id="factorialResult"></p>
    <p id="fibonacciResult"></p>
    <p id="primeResult"></p>
    <p id="palindromeResult"></p>
</body>
</html>
```

Output:



Aim :

Design a HTML having a text box and four buttons named Factorial, Fibonacci, Prime, and Palindrome. When a button is pressed an appropriate function should be called to display

1. Factorial of that number
2. Fibonacci series up to that number
3. Prime numbers up to that number
4. Is it palindrome or not

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Number Operations</title>
    <script>
        function factorial(n) {
            if (n < 0) return "Factorial not defined for
negative numbers";
            if (n === 0 || n === 1) return 1;
            let result = 1;
            for (let i = 2; i <= n; i++) {
                result *= i;
            }
            return result;
        }

        function fibonacci(n) {
            if (n < 1) return "Enter a positive integer";
            let fibSeries = [0, 1];
            while (true) {
                let next = fibSeries[fibSeries.length - 1] +
```

```
fibSeries[fibSeries.length - 2];
                if (next > n) break;
                fibSeries.push(next);
            }
            return fibSeries;
        }

        function primesUpTo(n) {
            if (n < 2) return "No prime numbers";
            let primes = [];
            for (let i = 2; i <= n; i++) {
                let isPrime = true;
                for (let j = 2; j * j <= i; j++) {
                    if (i % j === 0) {
                        isPrime = false;
                        break;
                    }
                }
                if (isPrime) primes.push(i);
            }
            return primes;
        }

        function isPalindrome(n) {
            let str = n.toString();
            let reversedStr = str.split('').reverse().join('');
            return str === reversedStr ? `${n} is a palindrome`
: `${n} is not a palindrome`;
        }

        function displayResult(type) {
            let num =
parseInt(document.getElementById("numberInput").value, 10);
            let result;
            switch(type) {
                case 'factorial':
                    result = `Factorial: ${factorial(num)}`;
                    break;
                case 'fibonacci':
                    result = `Fibonacci: ${fibonacci(num)}`;
                    break;
                case 'prime':
                    result = `Prime Numbers:
```

```
${primesUpTo(num)}`;
                    break;
                case 'palindrome':
                    result = isPalindrome(num);
                    break;
            }
            document.getElementById("result").textContent =
result;
        }
    </script>
</head>
<body>
    <h1>Number Operations</h1>
    <input type="number" id="numberInput" placeholder="Enter a
number">
    <button
onclick="displayResult('factorial')">Factorial</button>
    <button
onclick="displayResult('fibonacci')">Fibonacci</button>
    <button onclick="displayResult('prime')">Prime</button>
    <button
onclick="displayResult('palindrome')">Palindrome</button>
    <p id="result"></p>
</body>
</html>
```

Output:

Aim :

Write a program to validate the following fields in a registration page

1. Name (start with alphabet and followed by alphanumeric and the length should not be less than 6 characters)
2. Mobile (only numbers and length 10 digits)
3. E-mail (should contain format like xxxxxxx@xxxxxx.xxx)

File name: index.html

File Name :

index.html

Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Registration Form</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            margin: 50px;
        }
        .error {
            color: red;
            font-size: 12px;
        }
    </style>
</head>
<body>

    <h2>Registration Form</h2>
    <form id="registrationForm">
        <label for="name">Name:</label>
        <input type="text" id="name" placeholder="Enter your
name" required>
```

```
        <span class="error" id="nameError"></span>
        <br><br>

        <label for="mobile">Mobile:</label>
        <input type="text" id="mobile" placeholder="Enter your
mobile number" required>
        <span class="error" id="mobileError"></span>
        <br><br>

        <label for="email">Email:</label>
        <input type="email" id="email" placeholder="Enter your
email" required>
        <span class="error" id="emailError"></span>
        <br><br>

        <button type="button" id="register">Register</button>
    </form>

    <script>

document.getElementById("register").addEventListener("click",
function () {
            let name = document.getElementById("name").value;
            let mobile =
document.getElementById("mobile").value;
            let email = document.getElementById("email").value;

            let namePattern = /^[A-Za-z][A-Za-z0-9]{5,}$/;
            let mobilePattern = /^[0-9]{10}$/;
            let emailPattern = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-
9.-]+\.[a-zA-Z]{2,}$/;

            let valid = true;

            if (!namePattern.test(name)) {
                document.getElementById("nameError").innerText
= "Invalid name (must start with a letter, min 6 characters)";
                valid = false;
            } else {
                document.getElementById("nameError").innerText
= "";
            }
```

```
            if (!mobilePattern.test(mobile)) {

document.getElementById("mobileError").innerText = "Invalid
mobile number (10 digits only)";
                valid = false;
            } else {

document.getElementById("mobileError").innerText = "";
            }

            if (!emailPattern.test(email)) {
                document.getElementById("emailError").innerText
= "Invalid email format";
                valid = false;
            } else {
                document.getElementById("emailError").innerText
= "";
            }

            if (valid) {
                alert("Registration successful!");
            }
        });
    </script>

</body>
</html>
```

Output: